IES | GENAI FOR ENGINEERS

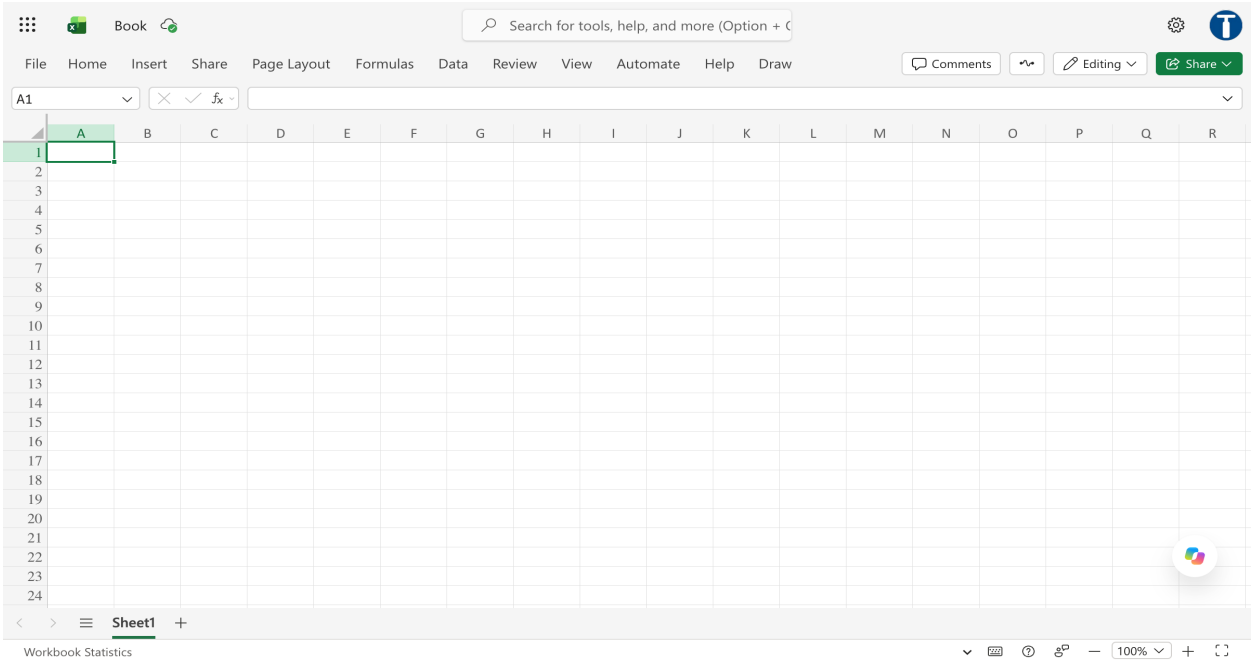
Lab 2 - Use Copilot in Excel to create an FMEA review

20 minutes | Synthetic classroom case

The team is reviewing four possible pump-system failures before the next operating campaign. Ratings are supplied classroom judgements, not field evidence.

1	Open Starter.xlsx and select a cell in FMEAInput.
2	Open Copilot in Excel.
3	Copy the prompt from the Copilot_Prompt sheet.
4	Ask Copilot to create the output and show its formulas.
5	Check one result yourself; correct the draft if needed.

Where to find Copilot



One prompt, one check

Paste this into Copilot in Excel

Act as an FMEA facilitator. Use only the named Excel table FMEAInput and preserve every row and supplied rating. Create an FMEA output table, calculate RPN = Severity x Occurrence x Detection, rank the four rows from highest to lowest RPN, and flag any row with Severity 9 or higher. For each row, propose one evidence request and one practical follow-up action. Do not invent incidents, change ratings, or approve the design. Show the RPN formula.

Check one result yourself

Verify FM01: 7 x 5 x 4 = 140. Also verify that FM04 is flagged because Severity = 9 even though its RPN is lower.

Expected Copilot draft

Four preserved failure modes, formula-based RPN values, a ranked priority, a severity flag, and evidence-focused actions.

What to submit

Save the generated output, your independent check, and one sentence stating what the result does not prove.

Microsoft 365 Copilot links

Microsoft 365 Copilot

Get started with Copilot in Excel

How Copilot Chat works in Microsoft 365 apps